

## Homework 3

There are a total of 5 questions.

**Q1 (Reflection):** Read the solution to HW 2.

- Are your own answers in line with the solutions? If not, list the questions you missed.
- Discuss what you could have done better.

**Q2 (Review question)** In this question, consider the following linear model,

$$y_i = x_i\beta + u_i, \quad i = 1, 2, \dots, N,$$

where  $y_i$  is a scalar and  $x_i$  is a  $k \times 1$  row vector. Define an estimator for  $\beta$  as

$$\hat{\beta} = \left( \sum_{i=1}^N x_i' x_i \right)^{-1} \left( \sum_{i=1}^N x_i' y_i \right)$$

Assume OLS1 and OLS2. In addition, assume that  $(y_i, x_i)$  is i.i.d. over  $i$ . Answer the questions below. All answers are in your notes. Try to understand (and convince yourself about) every step of the proof. When answering the questions below, be sure to highlight where the stated assumptions are used. If you have any questions or doubts, try to come to my office hours and ask questions.

1. Show that  $\hat{\beta}$  is a consistent estimator.
2. Derive the asymptotic distribution of  $\hat{\beta}$ .
3. Present a consistent estimator of the asymptotic variance of  $\hat{\beta}$ .
4. What is the standard error of  $\hat{\beta}$ ?

**Q3 (Data question)** Review the lab session notes 02 ([lab02.pdf](#)). Using the same data set (CPS85.txt), answer the following questions.

1. Estimate the following model that is similar to the one considered in the note,

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{exper}_i^2 + u_i$$

2. Compute a Q-Q plot (normal quantile plot) as in the lab notes. Read the appendix to Notes 03 that discusses the Q-Q plot. Do you think that  $u_i$  is normally distributed or not? If it is not normally distributed, is that a concern for your later regression analysis?

3. We consider the return to education. Let's increase years of education by 1. Thus, also years of experience will decrease by 1. The expected percentage change in the wage corresponding to this change,  $\theta$ , is given by

$$\left(1 + \frac{\theta}{100}\right) = E[\text{wage}|\text{educ}+1, \text{exper}-1] - E[\text{wage}|\text{educ}, \text{exper}] = \exp(\beta_1 - \beta_2 + (1 - 2\text{exper})\beta_3)$$

so that

$$\theta = 100 [\exp(\beta_1 - \beta_2 + (1 - 2\text{exper})\beta_3) - 1]$$

4. Provide an estimator for  $\theta$  (call it  $\hat{\theta}$ ). Argue that this estimator is a consistent estimator for  $\theta$ .
5. Note that this estimator depends on  $\text{exper}$ . Compute  $\hat{\theta}$  for values of  $\text{exper}$  ranging from 0 to 55 in steps of 5. Discuss your results.
6. Discuss how you would compute standard errors and confidence intervals for  $\hat{\theta}$  for values of  $\text{exper}$  ranging from 0 to 55. At this stage, you don't have to calculate them. Just discuss how you would do it.
7. (Optional; will not be graded) Compute standard errors and 95% confidence intervals for  $\hat{\theta}$  for values of  $\text{exper}$  ranging from 0 to 55 in steps of 5.

**Q4 (Monte Carlo Simulation and coverage rate)** Suppose we have a population model as

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$$

Because we fully control this environment, we can choose specifics of the population. In particular, we let

$$\beta_0 = 1, \beta_1 = 1, \beta_2 = 0.2$$

and

$$u \sim t_7$$

where  $t_7$  refers to  $t$ -distribution with degrees of freedom 7. Lastly, we assume that  $x$  follows normal distribution:

$$x \sim N(1, 4)$$

1. Randomly sample  $y$  and  $x$  from the population  $N$  times. Collect them in a vector

$$Y_N^{(1)} = [y_1, y_2, \dots, y_N]$$

and

$$X_N^{(1)} = [x_1, x_2, \dots, x_N]$$

$Y_N^{(m)}$  refers to a vector of length  $N$  randomly sampled from the population of  $y$ . The superscript denotes the  $m$ -th data set. Let  $N = 100$ .

2. Compute the OLS estimate for  $\beta = [\beta_0, \beta_1, \beta_2]'$  using the data set created in the previous question.
3. Compute the 95% confidence interval for  $\beta_1$  using the data set created in the previous question based on the asymptotic distribution you derived in Q2-2 (that is, the asymptotic distribution based on OLS1 and OLS2). Define an indicator variable

$$I^{(m)} = 1\{\beta_1 \text{ is contained in the 95\% confidence interval for } \beta_1\}$$

That is,  $I^{(m)}$  is one if the true value of  $\beta_1$  is contained in the 95% confidence interval for  $\beta_1$  using the  $m$ -th data set.

4. Repeat 1,2,3 for  $M$  times where  $M = 10,000$ . Compute the following quantity:

$$\text{empirical coverage rate} = \frac{1}{M} \sum_{m=1}^M I^{(m)}.$$

Does your answer meet your expectation? Explain why it meets or does not meet your expectations by checking the assumptions we learned in class.

5. Repeat 1,2,3,4 with the following modified model,

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \frac{|x|}{2} u$$

Note that we multiply  $\frac{|x|}{2}$  to  $u$ . a) What is the empirical coverage rate in this case? b) Compare it with the empirical coverage rate you obtained from the previous question. c) Does your answer meet your expectation? Explain why it meets or does not meet your expectations by checking the assumptions we learned in class.

**Q5 (Paper question):** Find a paper that performs empirical analysis based on the OLS estimation in your field. If you can't find it, then find such a paper from "American Economic Review", which is one of the premier journals in economics.

1. Find an academic paper<sup>1</sup> that (1) was published in one of those journals from your field of interest, AND (2) contains the word OLS or ordinary least squares, AND (3) reports empirical results based on the OLS estimation.

One way to find such a paper is to use Google Scholar. Type the following in the search box

`source:"[name of the journal]" "OLS" OR "ordinary least squares"`

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<sup>1</sup>If you can't find such a paper from the field of your interest, then find it from the "American Economic Review", which is one of the premier journals in economics.

2. Properly cite the paper you found (name of the author, the title of the article, year of publication, the name of the journal, etc.)
3. Read the paper and explain what is the main research question of the paper in one paragraph.
4. Let us write their empirical model using our notation

$$y_i = x_i\beta + u_i$$

What is  $y_i$  in the paper? What is  $x_i$  in their main specification? (That is, if there are several models with different choices of explanatory variables or dependent variable, just pick one) How many variables are included in  $x_i$ ? What is the total number of observations?

5. Do the authors state underlying assumptions needed for their OLS estimation and inference to be valid?
6. Argue convincingly that  $E[x_i u_i] = 0$  (exogeneity assumption) does not hold in their empirical application. That is, provide a “plausible” story (or, a counter example) that  $E[u_i x_i] = 0$  does not hold. Write your answer in one paragraph (say, no more than 5 sentences). Try to be as critical as you can.